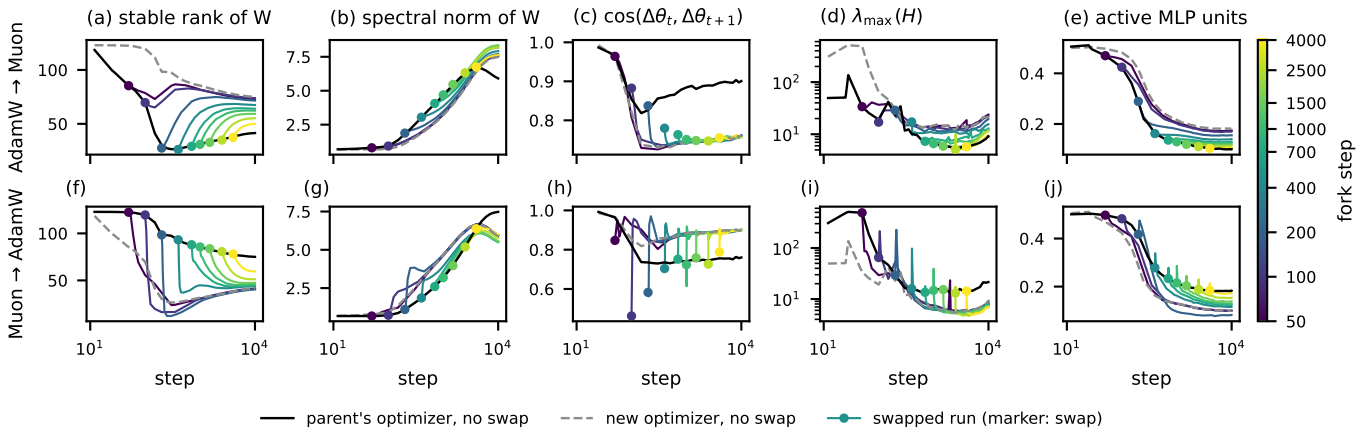

(a) stable rank of W
(b) spectral norm of W
(c) cos(Δθ_t, Δθ_{t+1})
(d) λ_max(H)
(e) active MLP units
AdamW → Muon
Muon → AdamW
(f)
(g)
(h)
(i)
(j)
step
fork step
4000
2500
1500
1000
700
400
200
100
50
parent's optimizer, no swap
new optimizer, no swap
swapped run (marker: swap)